

Weight Sensitivity and Seed-Block Resampling

Local robustness of the five-criterion TOPSIS

This supplement documents only the sensitivity reported in the current manuscript: local $\pm 20\%$ weight perturbations and RunID/seed-block bootstrap. It does not assign percentages or simulation counts that the paper does not report.

Study	9 methods $\times$ 4 scenarios $\times$ 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Objective

TOPSIS produces an ordering conditional on criteria, transformations, normalization, and weights. Sensitivity examines whether B4's nominal lead changes under reasonable local weight variations or resampling of the ten matched blocks.

2. Local weight perturbation

$$w'_j = w_j(1 + \varepsilon_j), \varepsilon_j \in [-0.20, +0.20]; \hat{w}_j = w'_j / \sum w'_k$$

- Start from a nominal five-criterion profile.
- Perturb every component within $\pm 20\%$.
- Renormalize so that $\sum \hat{w}_j = 1$.
- Recompute the weighted matrix, ideals, distances, CC, and rank.
- Record whether the first-ranked method changes.

3. Seed-block bootstrap

Resampling operates on RunID/seed blocks rather than disconnected individual observations. Each selected block contributes its Method $\times$ Scenario cells jointly, preserving paired structure. Means, Usafe, and equal-weight TOPSIS are recomputed for each resample.

4. Documented result

Check	Input	Result	Interpretation
Weight profiles	Equal, W1', W2', W3'	B4 rank 1	Stable across the four reported profiles
Local weights	$\pm 20\%$ perturbations	B4 remains rank 1	Local, not global, robustness
Seed blocks	Matched-block bootstrap	B4 remains rank 1 under equal weights	Sampling stability for this criterion set

The manuscript text does not specify the number of perturbation draws, the number of bootstrap resamples, or percentage leadership frequencies. For traceability, this supplement does not invent those values. They should be added only when archived scripts and outputs support them.

5. What the sensitivity analysis shows

- B4 is stable near the evaluated weight configurations.
- First rank does not depend on only one nominal combination.
- The result remains conditional on [Usafe, TP, T, Nstop, σv] and its normalization.

6. What it does not show

- It does not demonstrate universal B4 optimality.
- It does not replace weights elicited from operational requirements or experts.
- It does not prove robustness to different criteria, utility thresholds, or unsimulated physical conditions.